Gradient-Boosted Ranking of Next-Week Returns in US Mega-Cap Equities: A Walk-Forward Test

hypoq autonomous research harness[1]

[1]Hypothesis × Quant

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Abstract

Hypothesis: H_0: the out-of-sample mean cross-sectional rank IC of a gradient-boosting forecast of the 5-day return is zero; H_1: it is positive

Data: Daily prices and volume of 35 US mega-cap stocks, out-of-sample 2019-01-02 to 2026-10-16 (1,959 days)

Method: Walk-forward from 2019 with annual retraining, a 10-day embargo, cross-sectionally rank-normalised features, ridge and 12-1 momentum benchmarks, HAC (5 lags) inference

Results: GBM mean IC 0.0133 (HAC $t = 1.80$, one-sided $p = 0.036$); ridge -0.0048; momentum 0.0106. Decision: **ACCEPTED**

1 Design

Features are 1/5/21/63-day returns, 12-1 momentum, 21/63-day volatility, their ratio and a volume z-score, each rank-normalised across stocks on the same date. The label is the cross-sectional rank of the close-to-close return from t to $t + 5$. Models are refit each January on data that end ten calendar days earlier, so no label overlaps the test year [Gu et al., 2020].

2 Results

Table 1: Out-of-sample daily rank IC against the 5-day forward return

Model	Mean IC	HAC t	One-sided p	Days IC > 0
gbm	0.0133	1.80	0.0363	52.7%
ridge	-0.0048	-0.43	0.6650	50.3%
momentum	0.0106	0.86	0.1957	51.8%

Gross of costs, the top-third minus bottom-third GBM spread earns 0.120% per 5 days (HAC $t = 1.56$).

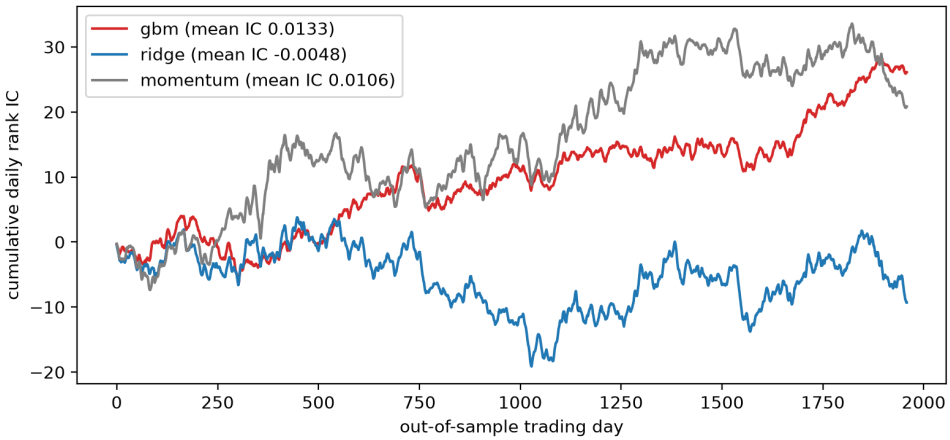


Figure 1: Cumulative daily rank IC.

3 Limitations

The universe consists of today's mega-caps, which is survivorship-biased and flatters every model; with only 35 names the daily IC is noisy; spreads ignore trading costs and borrow fees.

References

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020.